

10/9/23

## MATHS

YOUVA

Date:

$$36 = 2^2 \times 3^2$$

$$45 = 3^2 \times 5$$

1.  $\text{LCM}(36, 45) = 2^2 \times 3^2 \times 5 = 180 \text{ sec} = 3 \text{ min}$

$$36 = 2^2 \times 3^2$$

$$45 = 3^2 \times 5$$

(3) 3

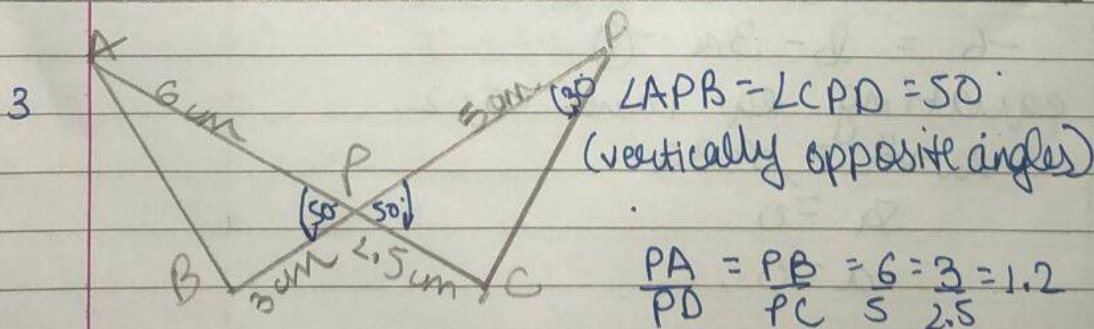
2.  $\frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ} \Rightarrow \frac{2 \left( \frac{1}{\sqrt{3}} \right)}{1 - \left( \frac{1}{\sqrt{3}} \right)^2} = \frac{\frac{2}{\sqrt{3}}}{\frac{3-1}{3}}$

$$= \frac{2/\sqrt{3}}{2/3}$$

$$= \frac{1}{\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}}$$

$$= \frac{3}{\sqrt{3}} = \sqrt{3}$$

(3)  $\tan 60^\circ$ 

by SAS similarity  $\triangle APB \sim \triangle DPC$   
 $\angle PAB = \angle PDC = 30^\circ$

$$\angle PBA = 180 - (50 + 30) = 180 - 80 = 100^\circ \quad (4) 100^\circ$$

4

$$\triangle ABC \sim \triangle QRP$$

$$\frac{ac(ABC)}{ac(PQR)} = \frac{9}{4} = \frac{BC^2}{PR^2}$$

$$\frac{9}{4} = \frac{(15)^2}{(PR)^2}$$

$$\frac{9}{4} = \frac{225}{PR^2}$$

$$PR^2 = 100$$

$$PR = 10 \text{ cm}$$

(1) 10cm

5

$$p(x) = x^2 + (a+1)x + b$$

$$p(2) = (2)^2 + (a+1)(2) + b$$

$$0 = 4 + 2a + 2 + b$$

$$0 = 6 + 2a + b$$

$$(-6) = 2a + b \quad \text{--- (1)}$$

$$p(-3) = (-3)^2 + (a+1)(-3) + b$$

$$0 = 9 - 3a - 3 + b$$

$$-6 = b - 3a \quad \text{--- (2)}$$

equating eq 1 and eq 2

$$2a + b = b - 3a$$

$$5a = 0$$

$$a = 0$$

6

$$\frac{14587}{2 \times 5^4}$$

$$2 \times 5^4$$

$$\text{denominator} = 2^m \times 5^n$$

$$2 \times 5^4 = 2^m \times 5^4$$

$$m = 1$$

$$n = 4 \quad \because m < n$$

the no. will terminate after 4 decimal places

(4) four decimal places

5

$$x = a$$

$$y = b$$

$$x - y = 2$$

$$a - b = 2 \quad \text{--- (1)}$$

$$x + y = 4$$

$$a + b = 4 \quad \text{--- (2)}$$

add eq 1 and eq 2

$$a - b + a + b = 2 + 4$$

$$2a = 6$$

$$a = 3$$

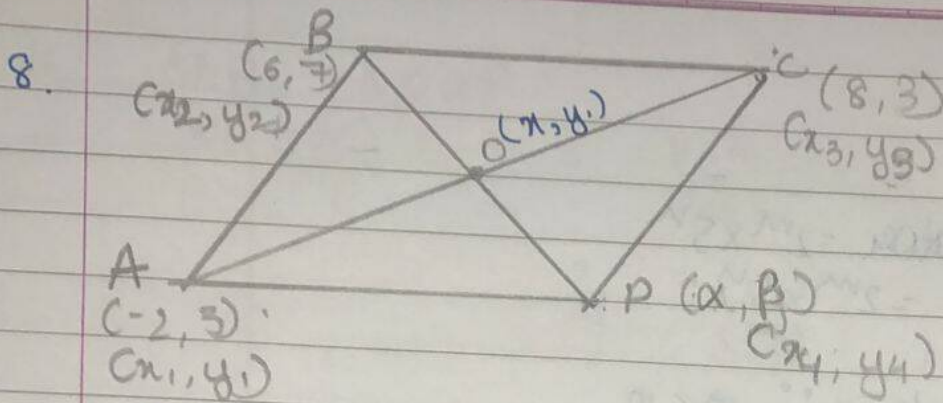
Put a in eq 1

$$3 - 2 = b$$

$$1 = b$$

(3) 3 and 1





Join AC and BD and intersect at O.

∵ diagonals bisect each other in a parallelogram (O is mid point of both the diagonals).

$$x = \frac{x_2 + x_4}{2} = \frac{x_1 + x_3}{2}$$

$$\left[ \frac{6 + x_4}{2} = \right]$$

$$\frac{6 + x}{2} = \frac{(-2) + 8}{2}$$

$$6 + x = 6$$

$$x = 0$$

$$y = \frac{y_2 + y_4}{2} = \frac{y_3 + y_1}{2}$$

$$\frac{7 + \beta}{2} = \frac{3 + 3}{2}$$

$$7 + \beta = 6$$

$$\beta = -1$$

$$\boxed{(0, -1)}$$

$$\boxed{\text{Ans: } (0, -1)}$$

9.

$$9x + 3y + 12 = 0$$

$$18x + 6y + 26 = 0$$

$$\left[ \begin{array}{ccc} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{array} \right] \frac{1}{2} \frac{9}{18} = \frac{3}{6} \neq \frac{12}{26}$$

$$\frac{1}{2} = \frac{1}{2} \neq \frac{6}{13}$$

(4) no solution

10

$$2x - 3y + 10 = 0$$

$$3x + ky + 15 = 0$$

as coincident lines

$$\frac{2}{3} = \frac{(-3)}{k} = \frac{10}{15}$$

$$\frac{2}{3} = \frac{(-3)}{k}$$

$$k = \left( \frac{-9}{2} \right)$$

(1)  $k = \left( \frac{-9}{2} \right)$

11

$$10x - 5y + 15 = 0$$

$$6x + ky + 19 = 0$$

as intersecting lines

$$\frac{5}{3} \neq \frac{(-5)}{k}$$

$$k \neq -3$$

(1)  $k \neq (-3)$



12 HCF (6, 8, 32) =  $2^3 = 8$

$616 = 2^3 \times 7 \times 11$   
 $32 = 2^5$

$\therefore$  8 divides  
 (1) 8

13 (1) similar

14  $a = x^3 y^4 z^2$   
 $b = x^2 y^2 z^4$

$LCM(a, b) = x^3 y^4 z^4$

(1)  $x^3 y^4 z^4$

15  $\alpha = (-3)$   
 $\beta = (4)$

$\alpha + \beta = -3 + 4 = 1$

$\alpha\beta = (-3)4 = (-12)$   
 $R[x^2 - x(\alpha + \beta) + \alpha\beta]$

$R[x^2 - x(-12) + (-12)]$   
 $R[x^2 - x - 12]$   
 $R\left[\frac{x^2 - x - 6}{2}\right]$

and: (3)  $\frac{x^2 - x - 6}{2}$

16 (3) interesting or coincident

17 (1) parallel

$$\begin{array}{r} 2616 \\ 2 \overline{) 308} \\ \underline{21} \phantom{154} \\ 7154 \\ \underline{71} \phantom{11} \\ 1 \end{array}$$

18

Let  $\Delta ABC \sim \Delta PQR$   
 $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$   
 $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = \frac{4}{8} = \frac{1}{2}$

(4) 16.81

19

A:  $x^2 - 4x - 5$

$x^2 - 5x + x - 5 = 0$   
 $x(x - 5) + (x - 5) = 0$   
 $(x + 1)(x - 5) = 0$   
 $x = 5, (-1)$

R:  $\alpha = 2 + \sqrt{3}$   
 $\beta = 2 - \sqrt{3}$

$\alpha + \beta = 4$

$\alpha\beta = (4 - 3) = 1$

$P(x) = x^2 - 4x + 1$

(3) Assertion (A) is true but Reason R is false

20  $3(R-1)x + 4y - 24 = 0$   
 $15x + 20y - 8(R+13) = 0$

$$\frac{3(R-1)}{18} = \frac{4}{20} = \frac{1}{5}$$

$$R-1 = 1$$

$$R = 2$$

(1) Both A and R are true and R is correct explanation of A.



To prove:

$6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$  is a composite no.

$$725 = 5 \times 5 \times 29 \times 1$$

$$\begin{array}{r|l} 5 & 725 \\ 5 & 145 \\ 29 & 29 \\ & 1 \end{array}$$

$\therefore$  it has more than two factors the number is composite number.  
 Hence Proved.

22

$$4x + 6y = 7 \Rightarrow 4x + 6y - 7 = 0$$

$$Rx + 18y = 21 \Rightarrow Rx + 18y - 21 = 0$$

$$a_1 = 4$$

$$b_1 = 6$$

$$c_1 = (-7)$$

$$a_2 = R$$

$$b_2 = 18$$

$$c_2 = (-21)$$

As a system of linear equation of infinite solutions.

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{4}{R} = \frac{6}{18}$$

$$\frac{4}{R} = \frac{1}{3}$$

$$R = 12$$

$\therefore R = 12$

23

Let distance between origin and  $(-6, 8)$  be  $AB$

$$x_1 = 0$$

$$y_1 = 0$$

$$x_2 = (-6)$$

$$y_2 = 8$$

Applying distance formula

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = AB$$

$$\sqrt{(-6)^2 + (8)^2} = AB$$

$$\sqrt{36 + 64} = AB$$

$$100 = AB$$

$$AB = 100$$

$\therefore$  distance between origin and  $(-6, 8)$

$$is 100$$

24.

$$\sin 3A = \cos(A - 10)$$

$$\sin A = \cos(90 - A)$$

$$so \sin 3A = \cos(90 - 3A)$$

$$\cos(90 - 3A) = \cos(A - 10)$$

on comparing angles

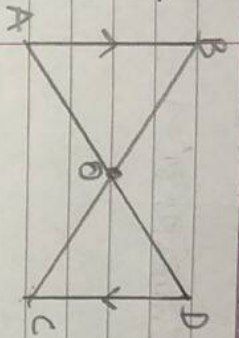
$$90 - 3A = A - 10$$

$$100 = 4A$$

$$25 = A$$

$\therefore A$  is equal to 25.

25.



given  $AB \parallel CD$

to prove  $\triangle AOB \sim \triangle DOC$

proof.

$$AB \parallel CD$$

$\angle ABO = \angle DCO$  [alternate interior angles]

$\angle BAO = \angle CDO$  [alternate interior angles]

by AA similarity criterion

$$\triangle AOB \sim \triangle DOC$$

Hence proved.



Topic: To prove:

26.  $\sin(50^\circ + \theta) - \cos(40^\circ - \theta) + \tan 1^\circ \cdot \tan 10^\circ \cdot \tan 80^\circ \cdot \tan 89^\circ = 1$

proof  $\sin(50^\circ + \theta) = \cos(90^\circ - (50^\circ + \theta))$   
 $= \cos(40^\circ - \theta)$

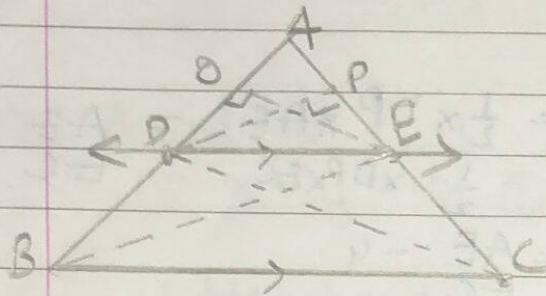
$\tan 1^\circ = \cot(90^\circ - 1^\circ)$   
 $= \cot 89^\circ$

$\tan 10^\circ = \cot(90^\circ - 10^\circ)$   
 $= \cot 80^\circ$

$\sin(50^\circ + \theta) - \cos(40^\circ - \theta) + \tan 1^\circ \cdot \tan 10^\circ \cdot \tan 80^\circ \cdot \tan 89^\circ$   
 $\cos(40^\circ - \theta) - \cos(40^\circ - \theta) + \cot 89^\circ \cdot \tan 89^\circ \cdot \cot 80^\circ \cdot \tan 80^\circ$   
 $\frac{0 + (1)(1)}{0 + 1} \left[ \because \tan \theta \times \cot \theta = \tan \theta \times \frac{1}{\tan \theta} = 1 \right]$   
 $1$

Hence proved.

27 Basic proportionality theorem states that, if a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points then other two sides are divided in same ratio.



Given  $ED \parallel BC$   
 To prove  $\frac{AD}{BD} = \frac{AE}{EC}$

Construct (1) Join BE and CD.

(2) then draw a perpendicular  $EO$  which is ~~perp~~ at  $90^\circ$  or such that  $AD \perp EO$ .

(3) draw a perpendicular DP such that  $DP \perp AE$



area of  $\triangle ADE = \frac{1}{2} \times AD \times ED$  - (1)

area of  $\triangle BDE = \frac{1}{2} \times BD \times ED$  - (2)

dividing eq 1 by eq 2  

$$\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times ED}{\frac{1}{2} \times BD \times ED} = \frac{AD}{BD}$$
 - (3)

Similarly,  $\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle CDE)} = \frac{\frac{1}{2} \times DP \times AE}{\frac{1}{2} \times DP \times EC} = \frac{AE}{EC}$   

$$\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle CDE)} = \frac{AE}{EC} - (4)$$

area  $[\triangle BDE] = \text{area}[\triangle CDE]$  - (5)  
 [  $\because$  they lie on same base DE and between 11<sup>th</sup> lines ]

from eq 4 and 5  

$$\frac{\text{area}(\triangle ADE)}{\text{area}(\triangle BDE)} = \frac{AE}{EC}$$
 - (6)

from eq 3 and eq 6  

$$\frac{AD}{BD} = \frac{AE}{EC}$$

Hence proved.

28

let, find change he receives extra change he receives (per day)

Case I: lotika paid ₹22.  
 $x + 4y = 22$  - (1)

Case II: Anand paid ₹16.  
 $x + 2y = 16$  - (2)

Subtract eq 2 from eq 1.  

$$(x + 4y) - (x + 2y) = 22 - 16$$
  

$$x + 4y - x - 2y = 6$$
  

$$2y = 6$$
  

$$y = 3$$

Substitute  $y = 3$  in eq 1.  
 $x + 4(3) = 22$   
 $x = 22 - 12$   
 $x = 10$

$\therefore$  find change is ₹10/- and extra change per day is ₹3/-

area of quadrilateral = area of  $\triangle ABC$  + area of  $\triangle ADC$

$$= \frac{35}{2} + \frac{109}{2}$$

$$= \frac{35+109}{2}$$

$$= \frac{144}{2}$$

$$= 72u^2$$

$\therefore$  area of quadrilateral is  $72u^2$

30. 1353 and 861.  
HCF (1353, 861) =

$$1353 = 861 \times 1 + 492$$

$$861 = 492 \times 1 + 369$$

$$492 = 369 \times 1 + 123$$

$$369 = 123 \times 3 + 0$$

$$\therefore 123 = 7 \times 17 + 2$$

=

30. HCF (1353, 861) =

$$1353 = 861 \times 1 + 492$$

$$861 = 492 \times 1 + 369$$

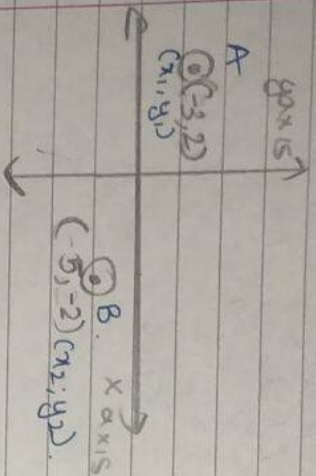
$$492 = 369 \times 1 + 123$$

$$369 = 123 \times 3 + 0$$

$\therefore$  HCF of 1352 and 861 is  $123$



31



$$A(-3, 2)$$

$$B(5, -2)$$

$$D(x, 0)$$

$$x_1 = (-3)$$

$$x_2 = 5$$

$$x_3 = (x)$$

$$AD = BD$$

Applying distance formula

$$\sqrt{(x_1 - x_3)^2 + (y_1 - y_3)^2} = \sqrt{(x_2 - x_3)^2 + (y_2 - y_3)^2}$$

Squaring on both the sides

$$(x_1 - x_3)^2 + (y_1 - y_3)^2 = (x_2 - x_3)^2 + (y_2 - y_3)^2$$

$$(-3 - x)^2 + (2 - 0)^2 = (5 - x)^2 + (-2 - 0)^2$$

$$\therefore (a - b)^2 = a^2 + b^2 - 2ab$$

$$(a + b)^2 = a^2 + b^2 + 2ab$$

$$9 + x^2 - 6x + 4 = 25 - 10x + 4$$

$$x^2 - 6x - 40 = 40$$

$$x^2 - 10x + 4x - 40 = 0$$

$$x(x - 10) + 4(x - 10) = 0$$

$$(x - 10)(x + 4) = 0$$

$$x - 10 = 0 \quad | \quad x + 4 = 0$$

$$x = 10 \quad | \quad x = -4$$

$\therefore$  points on x axis are  $(10)$  and  $(-4)$

32

$$\tan \theta + \cot \theta$$

$$\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$\frac{1 - \cos \theta}{\sin \theta} + \frac{1 + \sin \theta}{1 - \sin \theta}$$

$$\frac{\sin^2 \theta}{\cos \theta} + \frac{\cos^2 \theta}{\sin \theta}$$

$$\cos \theta (\sin \theta - \cos \theta) + \sin \theta (\cos \theta - \sin \theta)$$

$$\sin^2 \theta - \cos^2 \theta$$

$$\cos \theta (\sin \theta - \cos \theta) - \sin \theta (\cos \theta - \sin \theta)$$

$$\sin^2 \theta - \cos^2 \theta$$

$$\sin \theta \cos \theta (\sin \theta - \cos \theta)$$

$$(\sin \theta - \cos \theta) (\sin^2 \theta + \cos^2 \theta + \sin \theta \cos \theta) [a^3 - b^3 = (a - b)(a^2 + b^2 + ab)]$$

$$(\sin \theta \cos \theta) (\sin \theta - \cos \theta) [ \sin^2 \theta + \cos^2 \theta = 1 ]$$

$$1 + \sec \theta \csc \theta$$

$$1 + \sec \theta \csc \theta$$

$$1 + \sec \theta \csc \theta$$

$$1 + \sec \theta \csc \theta$$

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$$1 + \sec \theta \csc \theta$$

33

$$x - y + 1 = 0$$

$$2 - y + 1 = 0$$

$$3 = y$$

$$x - y + 1 = 0$$

$$x - y + 1 = 0$$

$$x - y + 1 = 0$$

$$x - y + 1 = 0$$

$$x - y + 1 = 0$$

$$x - y + 1 = 0$$

$$x = 0$$

$$y = 1$$

$$x = 1$$

$$y = 2$$

$$x = 2$$

$$y = 3$$

$$x = 3$$

$$y = 4$$

$$x = 4$$

$$x = 0$$

$$y = 1$$

$$x = 1$$

$$y = 2$$

$$x = 2$$

$$y = 3$$

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$$y = 4$$

$$x = 4$$

$$x = 0$$

$$y = 1$$

$$x = 1$$

$$y = 2$$

$$x = 2$$

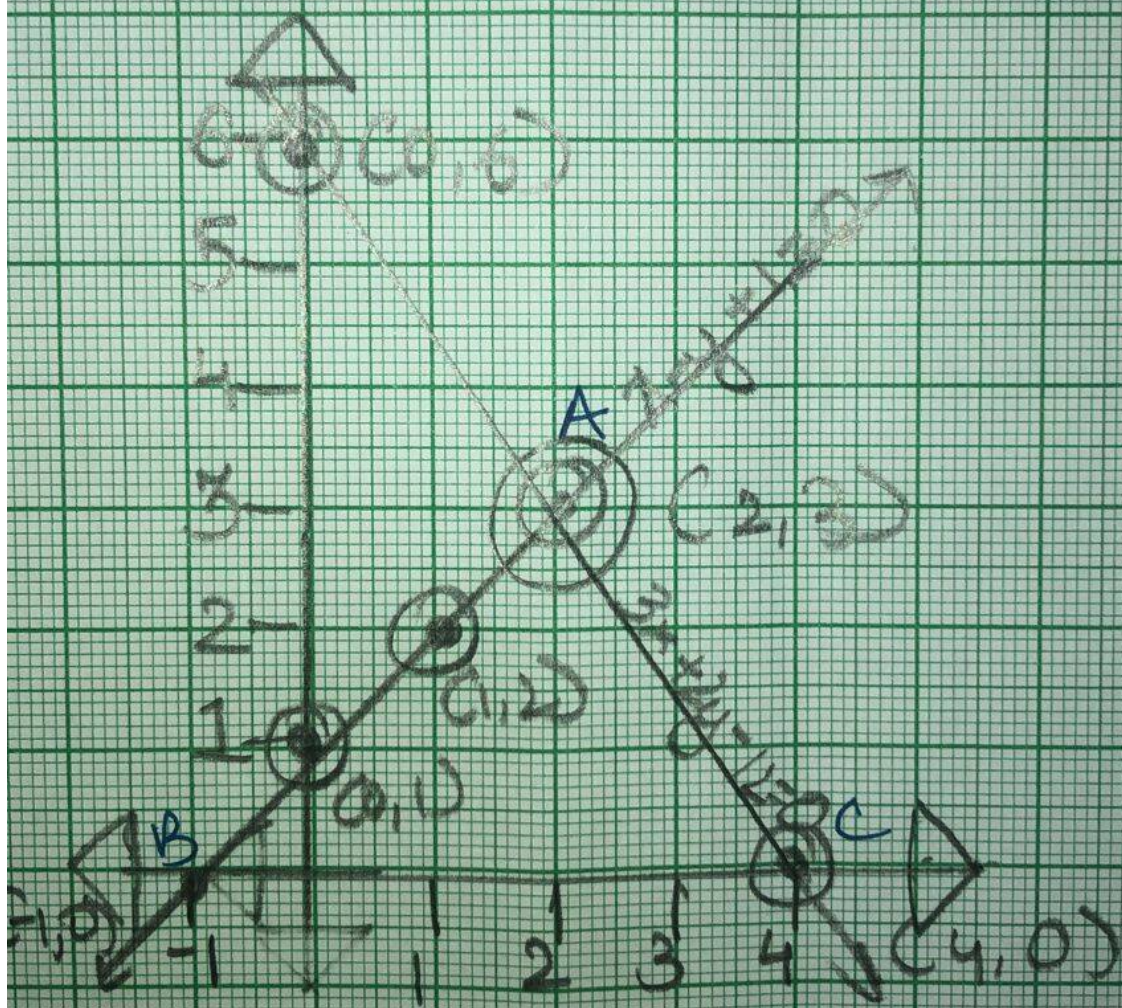
$$y = 3$$

$$x = 3$$

$$y = 4$$

$$x = 4$$







vertices of  $\Delta ABC$ .

A - (2, 3)

B - (-1, 0)

C - (4, 0)

$x_1 = 2$

$y_1 = 3$

$x_2 = -1$

$y_2 = 0$

$x_3 = 4$

$y_3 = 0$

area of  $\Delta ABC = \frac{1}{2} [x_1(y_2 - y_3)] + [x_2(y_3 - y_1)] + [x_3(y_1 - y_2)]$

$= \frac{1}{2} [2(0 - 0)] + [-1(0 - 3)] + [4(3 - 0)]$

$= \frac{1}{2} [0 + 3 + 12]$

$= \frac{15}{2} u^2$

$= 7.5 u^2$

$\therefore$  area of  $\Delta$   $7.5 u^2$



34.  $x = \sqrt{2}$   
 $x - \sqrt{2} = 0$   
 $x = -\sqrt{2}$   
 $x + \sqrt{2} = 0$

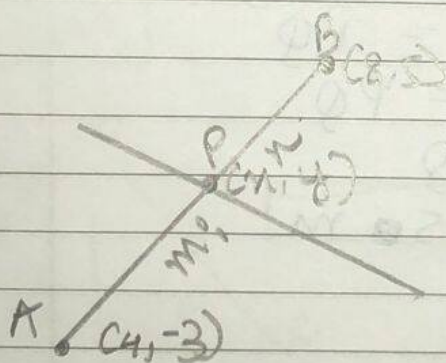
$$(x - \sqrt{2})(x + \sqrt{2}) = x^2 - 2 \quad [\because a^2 - b^2 = (a - b)(a + b)]$$

$$\begin{array}{r} 2x - 1 \\ x^2 - 2 \overline{) 2x^3 - 4x - x^2 + 2} \\ \underline{2x^3 - 4x} \phantom{- x^2 + 2} \\ 0 - x^2 + 2 \\ \underline{-x^2 + 2} \\ 0 \end{array}$$

So  $2x - 1 = 0$   
 $x = \frac{1}{2}$

all zeroes of polynomial  $2x^3 - 4x - x^2 + 2$  are  $\sqrt{2}$ ,  $-\sqrt{2}$  and  $\frac{1}{2}$

35.



$m \text{ of } AB = 3$   
 $m = 3$   
 $n = \frac{1}{3}$

$x_1 = 4$   
 $y_1 = -3$   
 $x_2 = 2$   
 $y_2 = 5$

$$\begin{aligned} x &= mx_2 + nx_1 \\ &= 3(2) + 1(4) \\ &= 6 + 4 \\ &= 10 \end{aligned}$$

$$\begin{aligned} y &= my_2 + ny_1 \\ &= 3(5) + 1(-3) \\ &= 15 - 3 \\ &= 12 \end{aligned}$$

$x = 10$

$y = 12$



coordinates of point P is  $(7, 3)$

36. (i) B  $(0, 12)$   
 Ans: (2)  $(0, 12)$

(ii) P  $(3, 5)$   
 Ans: (1)  $(3, 5)$

(iii) Q  $(6, 9)$  ~~not done~~  
 P  $(3, 5)$

$x_1 = 3$        $y_1 = 5$   
 $x_2 = 6$        $y_2 = 9$

Applying distance formula:

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = PQ$$

$$\sqrt{(3 - 6)^2 + (5 - 9)^2} = PQ$$

$$\sqrt{(-3)^2 + (-4)^2} = PQ$$

$$\sqrt{9 + 16} = PQ$$

$$\sqrt{25} = PQ$$

$$PQ = 5 \text{ m}$$

(3) 5m

37 (i) (3) parabola

(ii) (3) 0

(iii) ~~(3, 2)~~

Ans: (3)  $(-3, 2)$

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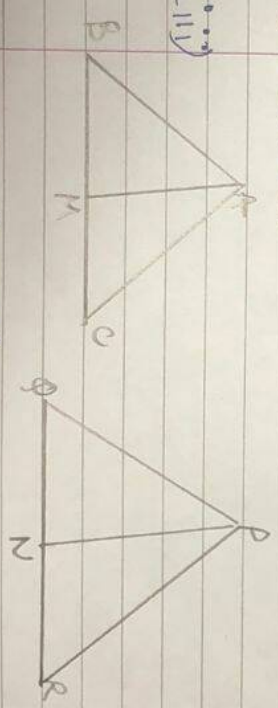
Area of  $\triangle PQR = 2 \times (\text{Area of } \triangle ABC)$

$$= 2 \left( \frac{1}{2} \times 6 \times 8 \right)$$

$$= 48 \text{ unit}$$

(i) (1) RHS

(ii)



$$\text{Area}(\triangle ABC) = \frac{AM^2}{2}$$

$$\text{Area}(\triangle PQR) = \frac{PN^2}{2}$$

$$\sqrt{\frac{4}{9}} = \frac{AM}{PN}$$

$$\frac{2}{3} = \frac{AM}{PN}$$

Ans: (1) 2:3